

Synthetic Uniform-Flow Generation and 2WCF Validation

Technical documentation for the current Python generator and validator

Scripts documented:

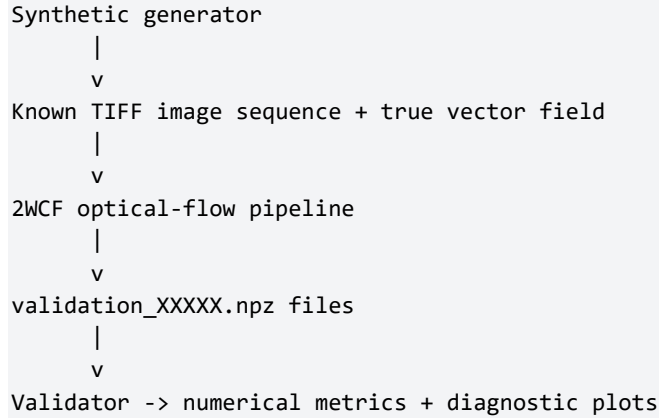
generate_uniform_flow_test.py
evaluate_2wcf_uniform_test.py

Primary purpose	Generate known synthetic motion and quantify 2WCF accuracy
Ground-truth units	pixels per frame (px/frame)
Coordinate convention	+u = right, +v = down

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1. System overview

The validation workflow creates a synthetic image sequence with a mathematically prescribed uniform displacement, processes that sequence with the existing 2WCF optical-flow pipeline, and compares the estimated vectors with the exact ground truth. The goal is not to replace the 2WCF algorithm, but to provide a controlled benchmark in which the correct answer is known.



2. File formats and their roles

Format	Produced by	Main contents	Role
.tiff	Generator	Synthetic particle images	Input image sequence for 2WCF
.npz	Generator	u_gt, v_gt, true_u, true_v, true_speed	Exact ground truth
.npz	2WCF	raw/processed/output vectors, masks, consistency, confidence	Internal validation diagnostics
.pkl / .wxf	2WCF	Final retained x, y, u, v vectors	Normal final optical-flow output
.csv / .txt / .png	Validator	Metrics, summary, plots	Human-readable validation results

The validation-specific .npz files are intentionally separate from the normal PKL/WXF output. PKL and WXF represent the final retained velocity field, while NPZ stores multiple NumPy arrays needed to inspect intermediate stages of the validation pipeline.

3. Synthetic uniform-flow generator

The generator creates a sequence in which every particle undergoes the same displacement per frame. This is the simplest possible optical-flow benchmark and is used to establish baseline accuracy before testing velocity gradients, vortices, noise, or SVD filtering.

3.1 User settings

```
WIDTH = 512
HEIGHT = 512
N_FRAMES = 100
TARGET_VISIBLE_PARTICLES = 4000
TRUE_U = 1
TRUE_V = 0
PARTICLE_SIGMA = 1.1
PARTICLE_PEAK = 220.0
SEED = 42
```

Parameter	Meaning	Current value
WIDTH, HEIGHT	Image dimensions in pixels	512 x 512
N_FRAMES	Number of generated images	100
TARGET_VISIBLE_PARTICLES	Approximate tracer count in the visible window	4000
TRUE_U	Exact horizontal displacement per frame	1 px/frame
TRUE_V	Exact vertical displacement per frame	0 px/frame
PARTICLE_SIGMA	Gaussian particle-image width	1.1 px
PARTICLE_PEAK	Approximate isolated-particle peak intensity	220
SEED	Random seed for reproducibility	42

Coordinate convention: +u points right and +v points down because the generator uses image coordinates.

3.2 Run folder naming

The current script creates the run folder using the prescribed velocity components:

```
RUN_NAME = f"synthetic_uniform_u{TRUE_U}_v{TRUE_V}"
```

For `TRUE_U = 1` and `TRUE_V = 0`, the folder is named `synthetic_uniform_u1_v0`. The helper function `format_value_for_filename()` is present in the script, but the current `RUN_NAME` line does not use it. Therefore decimal values currently appear with ordinary decimal points unless `RUN_NAME` is changed.

3.3 Particle rendering

Particle centers are stored as floating-point coordinates. The generator performs bilinear splatting, which distributes a particle between its four neighboring pixels according to its fractional position. A Gaussian blur is then applied to represent an optical point-spread function. This preserves subpixel motion and allows tests such as 0.25 px/frame or 0.5 px/frame.

```
floating-point particle center
-> bilinear splatting
-> Gaussian blur (sigma = PARTICLE_SIGMA)
-> fixed intensity scaling
-> uint8 TIFF image
```

The generator does not normalize every frame independently. A fixed Gaussian scaling factor is calculated once, which avoids introducing artificial frame-to-frame brightness changes.

3.4 Extended particle domain

Particles are seeded outside the visible image as well as inside it. This is necessary because uniform translation moves particles out of the camera field of view. Seeding an extended domain allows new particles to enter naturally, so the visible tracer density does not collapse over time.

```
total_dx = abs(TRUE_U) * (N_FRAMES - 1)
total_dy = abs(TRUE_V) * (N_FRAMES - 1)
margin_x = ceil(total_dx) + render_padding + 10
margin_y = ceil(total_dy) + render_padding + 10
```

3.5 Particle motion

For frame index t , each particle position is calculated analytically as:

```
x(t) = x_initial + TRUE_U * t
y(t) = y_initial + TRUE_V * t
```

The current uniform-flow generator has no acceleration, no spatial velocity gradient, no synthetic image noise, and no out-of-plane motion. Those effects should be added only after the baseline validation is understood.

3.6 Generator outputs

```
synthetic_uniform_u1_v0/
|-- frames/
|   |-- frame_00000.tiff
|   |-- frame_00001.tiff
|   `-- ...
|-- true_vector_values.npz
`-- parameters.json
```

3.7 Ground-truth NPZ

Array/value	Meaning	Shape/type
u_gt	True horizontal displacement at every pixel	HEIGHT x WIDTH float32
v_gt	True vertical displacement at every pixel	HEIGHT x WIDTH float32
true_u	Scalar prescribed horizontal displacement	float32
true_v	Scalar prescribed vertical displacement	float32
true_speed	$\sqrt{\text{true_u}^2 + \text{true_v}^2}$	float32

Current implementation note: The generator saves the ground-truth file as `true_vector_values.npz`, but its final print statement still refers to `ground_truth.npz`. The saved file itself is correct; the printed path should be updated if you want the console message to match the actual filename.

4. Interface with the 2WCF optical-flow pipeline

The generated `frames/` directory is used as the input image folder for the 2WCF script. The optical-flow calculation is performed normally. The validation output is additional diagnostic data and does not replace the standard PKL/WXF velocity files.

```

generator frames/
  -> 2WCF Farneback calculation
  -> forward/backward consistency
  -> optional processing/smoothing
  -> sampled validation arrays
  -> final retained velocity field

```

4.1 Validation arrays expected from 2WCF

Name	Meaning
x, y	Sample-grid coordinates
u_raw, v_raw	Raw Farneback flow before final rejection/smoothing
u_processed, v_processed	Processed flow before final output selection
u_output, v_output	Final output vector components on the sampled grid
valid_mask	Final retained/rejected vector mask
consistency_mask	Forward-backward consistency pass mask
consistency_error	Forward-backward consistency error
confidence	Confidence derived from consistency diagnostics

The validation density is controlled by the sampled grid used by the 2WCF pipeline. If ArrowDensityStep is 12, the validation arrays contain approximately one location every 12 pixels in x and y, after the dense Farneback field has already been computed.

5. 2WCF uniform-flow validator

The validator reads the synthetic ground truth and every validation_XXXXX.npz file created by the optical-flow run. It then compares the optical-flow vectors with the true vector at the same sampled coordinates.

5.1 Required paths

```

SYNTHETIC_TEST_DIR = Path(r".../synthetic_uniform_u1_v0")
PIPELINE_OUTPUT_DIR = Path(r".../frames_run_1")
GT_PATH = SYNTHETIC_TEST_DIR / "true_vector_values.npz"
VALIDATION_DIR = PIPELINE_OUTPUT_DIR / "validation"

```

The two directories must correspond to the same experiment. A common validation mistake is to process the frames from one synthetic run but point the validator to the ground truth from another run. That creates a consistent-looking but incorrect error report.

Recommended check: Before each validation run, confirm that the generator TRUE_U/TRUE_V values, the 2WCF image_folder, and SYNTHETIC_TEST_DIR all refer to the same dataset.

5.2 Ground-truth lookup

The optical-flow validation grid is sampled, so the validator does not compare against the entire full-resolution ground-truth field. Instead, it uses each sampled x/y location to look up the correct local ground-truth vector.

```

u_truth = u_gt_full[y, x]
v_truth = v_gt_full[y, x]

```

For uniform flow these values are identical everywhere. For future non-uniform tests, such as shear or a vortex, the same mechanism will retrieve the correct local vector automatically.

5.3 Interior-region analysis

The validator reports both full sampled-field retention and an interior-region result. The current interior border is 24 pixels. This prevents optical-flow boundary effects from dominating the primary accuracy statistics.

```
INTERIOR_BORDER_PX = 24
```

Boundary errors are expected because particles can enter or leave the field of view and Farneback neighborhoods are incomplete near image edges.

6. Validation metrics

6.1 Endpoint error (EPE)

Endpoint error measures the Euclidean distance between the estimated velocity vector and the true vector:

```
EPE = sqrt((u_est - u_true)^2 + (v_est - v_true)^2)
```

EPE is expressed in px/frame. A value of 0 means an exact estimate. The validator reports the median, mean, and 95th-percentile EPE for the final retained interior vectors.

6.2 Component bias

```
u_bias = u_est - u_true  
v_bias = v_est - v_true
```

A negative bias indicates systematic underestimation of that component; a positive bias indicates overestimation. Bias is useful because a low EPE can still hide a systematic directional tendency.

6.3 Relative endpoint error

```
relative_EPE[%] = 100 * median_EPE / true_speed
```

This expresses the typical vector error as a percentage of the prescribed flow magnitude. It should not be used when `true_speed = 0`.

6.4 Angular error

Angular error measures the angle between the estimated and true velocity vectors. It is reported in degrees. This metric is undefined for zero-magnitude vectors and is therefore represented as NaN where necessary.

6.5 Vector retention

```
retention[%] = 100 * retained_vectors / sampled_vectors
```

Retention must be interpreted together with accuracy. An algorithm could appear accurate by rejecting most difficult vectors. A good clean-flow result therefore combines low EPE with high retention.

6.6 Forward-backward consistency and confidence

The validator also reports the median forward-backward consistency error, the confidence value, and the fraction of interior vectors that pass the consistency test. These metrics assess internal self-consistency; they do not by themselves prove agreement with the ground truth.

7. Stage comparison

The validator separately calculates the median EPE of the raw Farneback field and the processed field. This is useful when the final result is poor because it shows whether the error was already present in Farneback or was introduced later by filtering or processing.

```
raw Farneback EPE
    vs
processed EPE
    vs
final retained-output EPE
```

8. Validator outputs

```
synthetic_validation_results/
|-- uniform_validation_per_frame.csv
|-- uniform_validation_summary.txt
|-- epe_vs_frame.png
|-- retention_vs_frame.png
|-- component_bias_vs_frame.png
`-- spatial_median_epe.png
```

Output	Purpose
uniform_validation_per_frame.csv	All calculated metrics for each frame pair
uniform_validation_summary.txt	Median summary across all frame pairs
epe_vs_frame.png	Median and 95th-percentile EPE over time
retention_vs_frame.png	Percentage of interior vectors retained over time
component_bias_vs_frame.png	Median u and v bias over time
spatial_median_epe.png	Median spatial EPE pattern across the sequence

9. How to interpret the diagnostic plots

9.1 EPE versus frame pair

- A low, nearly horizontal median-EPE curve indicates stable accuracy over time.
- The 95th percentile shows the upper tail of the error distribution without being dominated by the most extreme outliers.
- A growing error with frame index may indicate particle-density changes, edge effects, or a systematic temporal problem.

9.2 Retention versus frame pair

- Values close to 100% are expected for clean synthetic translation.
- Large drops in retention mean the consistency test is rejecting a significant fraction of sampled vectors.
- High accuracy should never be interpreted without checking retention at the same time.

9.3 Component bias

- Bias near zero means the corresponding velocity component is estimated without systematic offset.
- A stable negative line indicates systematic underestimation.
- A stable positive line indicates systematic overestimation.

9.4 Spatial median EPE

- A uniform low-error interior is the expected pattern for a good uniform-flow result.
- Higher error near image boundaries is expected because of finite neighborhoods and particles crossing the field of view.
- Strong internal structures or localized high-error regions should be investigated, especially if they are not associated with boundaries.

10. Recommended validation workflow

1. Choose TRUE_U and TRUE_V in the generator.
2. Generate the synthetic sequence and confirm the output folder.
3. Point the 2WCF image_folder to the matching frames/ directory.
4. Run 2WCF with validation data enabled.
5. Point SYNTHETIC_TEST_DIR and PIPELINE_OUTPUT_DIR in the validator to the matching run.
6. Run the validator and inspect the text summary before the plots.
7. Confirm low EPE, small bias, and high retention.
8. Repeat for a range of uniform displacements before moving to non-uniform flows.

11. Suggested next benchmark: uniform velocity sweep

After a single clean translation is validated, the next test should determine the displacement range over which the current 2WCF parameters remain accurate.

```
0.10, 0.25, 0.50, 1, 2, 4, 8, 12, 16, 20 px/frame
```

For each case, record measured speed, component bias, median EPE, 95th-percentile EPE, angular error, and retention. The most useful summary graph is measured velocity versus true velocity, with the ideal $y = x$ line shown for reference.

12. Known implementation notes

- The generator currently saves true_vector_values.npz, while its final print statement still names ground_truth.npz. The console print should be corrected for consistency.
- format_value_for_filename() exists but is not currently used in RUN_NAME. Decimal and negative run names therefore use the raw Python value representation unless this is changed.
- The validator assumes the sampled x/y grid is identical for every frame pair; it raises an error if the grid changes.
- The validator aggregates most summary metrics by taking the median across frame pairs.
- For a zero-speed ground truth, relative EPE and angular error need special interpretation because their normal denominators vanish.

13. Pre-run checklist

- ☐ Generator TRUE_U / TRUE_V are correct.
- ☐ Generator output folder name matches the intended case.

- ☐ 2WCF image_folder points to that exact frames/ directory.
- ☐ 2WCF validation output is enabled and writes validation_*.npz.
- ☐ Validator SYNTHETIC_TEST_DIR points to the same synthetic case.
- ☐ Validator PIPELINE_OUTPUT_DIR points to the 2WCF output created from those frames.
- ☐ true_vector_values.npz exists before validation is started.